Daimler Truck AG
Daimler Buses GmbH
Volvo Truck Corporation
MAN Truck & Bus SE
Scania AB
Scania CV
Volvo Bus Corporation
Renault Trucks
IVECO Group
DAF Trucks N.V.
VDL Bus & Coach B.V.
Ford Otosan

FMS-Standard description

Version 05

11.11.2025

© HDEI / BCEI Task Force

General annotations

- Data might be not available during ignition off / main switch off / charging *
- Most of the values are reliable after ca. 10 seconds after “ignition on”. *
- Physical Layer according to ISO 11898 (250kb/s)
- Application Layer according to SAE J1939/ 71 (J1939DA_202007)
- Data Link Layer according to SAE J1939/ 21
- If there is a discrepancy between definitions in this document and the SAE, the SAE documents are valid only except broadcast for
  * PGN 0x00FE6B (Driver ID)
  * PGN 0x00FE70 (Combination Vehicle Weight)
  * PGN 0x00FEE6 (Time/Date)
  * PGN 0x00FCB7 (Vehicle Electrical Power #4)
  * PGN 0x00FDC2 (Electronic Engine Controller 14)
  * PGN 0x00FEAF (Fuel Consumption (Gaseous))
  * PGN 0x00F009 (Vehicle Dynamic Stability Control 2)
  * PGN 0x00FEFC (Dash Display 1): Only used if different fuel types in different storage container
    Fuel Level 1 represents always the fuel used for driving without indication of the location of the storage container.
    Fuel Level 2 is not indicating the location of the storage container
  * PGN 0x00FC5E (High Voltage Energy Storage System History)
    This PGN is not sent as a BAM
  * PGN 0x00F009 (Vehicle Dynamic Stability Control 2) repetition rate is 100 ms instead of 10 ms
- The priority/source address of each OEM is different.
- If the information is delivered the function/data has to be delivered according to FMS-standard definition.

- The description of the connector(s) can be downloaded in the download area of FMS-Standard (Bus and Truck)
- If the information is not available, the function/data has to be sent as not available according to SAE
- “not used for (Bus) FMS-standard” means that there might be data sent according SAE but are not used in (Bus) FMS-standard interface. If no information is sent, then it has to be sent as “not available” (don’t care).
- “reserved for (Bus) FMS-standard” means that as long as there is no definition it is sent “FF (not available)”
- The accuracy/interpretation of signals might differ depending on vehicle brand/models *.
- (Bus) FMS-Standard is designed for Diesel engines. If it is used in vehicles with other engine types (e.g., dual engine) the information delivered might be different: *
- Some data are configuration relevant *

* Details can be obtained from the OEM

History

Truck	Bus
Changes / Addition to Vers. 01.00 Oct. 2009 • added history • change of DaimlerChrysler to Daimler • added Renault Trucks • update General Annotation • added description acc. SAE (based on Jan 2008 version) • deleted SAE ref as no longer valid • added additional comments • correction of PGNs (dez) in Example for BAM • added Priority to Example for BAM • added 2.2 Example SW Identification for buses and/or trucks • added Overview Messages	**Changes / Addition to Vers. 00.01:** • added Driver Identification DI • added Fuel Economy LFE • added Tell Tale Status TTS • added Example Tell Tale Status • added Overview Messages • added description acc. SAE J 1939 to the PGN´s • added additional comments to the PGN´s
Changes / Addition to Vers. 02.00 Sept. 2010 • update History • added 1.14 Ambient Conditions: AMB • added 1.15 Driver's Identification: DI • added 1.16 Fuel Economy: LFE • added 1.2 EEC2: Engine Percent Load At Current Speed • added 1.17 PTO Drive Engagement: PTODE • added 1.18 High Resolution Fuel Consumption (Liquid): HRLFC • update 3. Overview Messages •	**Changes / Addition to Vers. 00.02 dated 07.07.2009** • Page 36: Remark for accelerator position: "Daimler calculate from torque demand" deleted
Changes / Additions to Vers. 02.00 Nov. 2010 • some editorial corrections	

Version 03 dated 12.03.2012
One document for Bus and Truck FMS-Standard

Truck and Bus:

- update History
- update company names
- update General Annotation
- added SPN 513 Actual Engine – Percent Torque
- added High Resolution Fuel Consumption (Liquid) HRLFC
- added Aftertreatment 1 Diesel Exhaust Fluid Tank 1 Level
- added FMS Tell Tale Status : FMS1
- update example tell tale status
- additions to additional comments
- update Overview messages

Truck section:

- added Combination Vehicle Weight: CVW
- added Electronic Retarder Controller 1: ERC1
- added 1.22 Air Supply Pressure : AIR1
- update 3. Overview Messages

Version 03 dated 12.03.2012
One document for Bus and Truck FMS-Standard

Truck and Bus:

- update History
- update company names
- update General Annotation
- added SPN 513 Actual Engine – Percent Torque
- added High Resolution Fuel Consumption (Liquid) HRLFC
- added Aftertreatment 1 Diesel Exhaust Fluid Tank 1 Level
- added FMS Tell Tale Status : FMS1
- update example tell tale status
- additions to additional comments
- update Overview messages

Bus section:

- connector description in separate file

Version 03 dated 28.01.2016

Truck and Bus:

- update History
- correction of ISO No. for Tell Tale ID 53

Version 03 dated 28.01.2016

Truck and Bus:

- Update History
- correction of ISO No. for Tell Tale ID 53

Version 04 dated 13.10.2017	Version 04 dated 13.10.2017
Truck and Bus: • update History • added Electronic Brake Controller 1: EBC1 • added Electronic Engine Controller 14: EEC14 • added Fuel Consumption (Gaseous): GFC • added SPN 38 Fuel Level 2 • added Truck and Bus: Retarder Information : ERC1 (moved from Truck only) • added Tell Tale No 61-64 • update 3. Overview Messages • renamed PGN 0x00FEFC: Dash Display 1: DD1 **Truck section:** • added SPN 70 Parking Brake Switch • added SPN 527 Cruise Control States • renamed PGN 0x00FEF1: Cruise Control/Vehicle Speed 1: CCVS1	**Truck and Bus:** • update History • added Electronic Brake Controller 1: EBC1 • added Electronic Engine Controller 14: EEC14 • added Fuel Consumption (Gaseous): GFC • added SPN 38 Fuel Level 2 • added Truck and Bus: Retarder Information : ERC1 (moved from Truck only) • added Tell Tale no 61-64 • update 3. Overview Messages • renamed PGN 0x00FEFC: Dash Display 1: DD1 **Bus section:** • added SPN 527 Cruise Control States • added SPN5464 Hybrid Battery Pack Remaining Charge • added SPN 1807 Steering Wheel Angle

Version 04 dated 17.09.2021	Version 05 dated 24.11.2023
Truck and Bus: • update History • added Tell Tale No 65-69	**Truck and Bus:** • update History • added Vehicle Weight: VW (moved from Truck only) • added SPN 13171 DC Charging State • added SPN 15268 HVESS Estimated Remaining Distance • added SPN 241 Tire Pressure • added SPN 929 Tire Location • added SPN 8428 Fuel Supply Estimated Remaining Distance • added SPN 97 Water In Fuel Indicator 1 • Update SPN 5837 Fuel Type list **Bus section:** • added SPN 1810 Longitudinal Acceleration • added SPN 1809 Lateral Acceleration • added SPN 8235 HVESS Thermal Management System Mode • added SPN 7315 Propulsion System Active • added SPN 175 Engine Oil Temperature 1 • added SPN 100 Engine Oil Pressure 1 • added SPN 170 Cab Interior Temperature • added SPN 177 Transmission Oil Temperature 1 • added SPN 8211 HVESS Total Energy Throughput

Version 05 dated 07.07.2024	Version 05 dated 11.11.2025
Truck and Bus: • update History • update General annotations • additional Annotations PGN FD7D FMS1 • additional comment PGN FAB8 EVSE1DCS1 **Bus section** • error correction PGN 64606 HVESSHIST • additional comment PGN 64606	**Truck and Bus:** • update History • update General annotations • added Tell Tale No 70-75 • additional comment for Tell Tales **Bus section** •

1 PARAMETERS FOR FMS GATEWAY (ACCORDING TO SAE J1939) 11

1.1 Parameters for Bus and Truck FMS-Standard 11
1.1.1 Fuel Consumption: LFC 11
1.1.2 Dash Display 1: DD1 12
1.1.3 Electronic Engine Controller #1: EEC1 13
1.1.4 Engine Hours, Revolutions: HOURS 14
1.1.5 Vehicle Identification: VI 15
1.1.6 FMS-standard Interface Identity / Capabilities: FMS 16
1.1.7 High Resolution Vehicle Distance: VDHR 18
1.1.8 Tachograph : TCO1 19
1.1.9 Engine Temperature 1: ET1 21
1.1.10 Ambient Conditions: AMB 22
1.1.11 Driver's Identification: DI 23
1.1.12 Fuel Economy: LFE 24
1.1.13 Air Supply Pressure : AIR1 25
1.1.14 High Resolution Fuel Consumption (Liquid): HRLFC 26
1.1.15 Aftertreatment 1 Diesel Exhaust Fluid Tank 1 Information: AT1T1I 27
1.1.16 FMS Tell Tale Status: FMS1 28
1.1.17 Electronic Brake Controller 1: EBC1 32
1.1.18 Electronic Engine Controller 14: EEC14 33
1.1.19 Fuel Consumption (Gaseous): GFC 36
1.1.20 Electronic Retarder Controller 1: ERC1 37
1.1.21 Vehicle Weight: VW 38
1.1.22 EVSE 1 DC Status 1 : EVSE1DCS1 39
1.1.23 EV Operator Indicators 1 : EVOI1 40
1.1.24 Tire Condition Message 1 : TIRE1 41
1.1.25 Operator Indicators : OI 42

1.2 Parameters for Truck FMS-Standard 43
1.2.1 Cruise Control/Vehicle Speed 1: CCVS1 43
1.2.2 Electronic Engine Controller #2: EEC2 45
1.2.3 Service Information: SERV 46
1.2.4 PTO Drive Engagement: PTODE 47
1.2.5 Combination Vehicle Weight: CVW 48

1.3 Parameters for Bus FMS-Standard 49
1.3.1 Cruise Control/Vehicle Speed: CCVS 49
1.3.2 Electronic Engine Controller #2 : EEC2 51
1.3.3 Door Control 1: DC1 52
1.3.4 Door Control 2: DC2 53
1.3.5 Time / Date : TD 54
1.3.6 Alternator Speed : AS 55
1.3.7 Electronic Transmission Controller 2 : ETC2 56
1.3.8 Air Suspension Control 4 : ASC4 57
1.3.9 Vehicle Electrical Power #4 : VEP4 58
1.3.10 Vehicle Dynamic Stability Control 2 : VDC2 59
1.3.11 Engine Temperature 1: ET1 60
1.3.12 Ambient Conditions: AMB 61
1.3.13 High Voltage Energy Storage System Thermal Management System Status 1 : HVESSTS1 62
1.3.14 High Voltage Energy Storage System History : HVESSHIST 63
1.3.15 Hybrid System Status 1 : HSS1 64
1.3.16 Engine Fluid Level/Pressure 1 : EFL/P1 66
1.3.17 Transmission Fluids 1 : TRF1 67
2 EXAMPLES 68
2.1 Broadcast Announce Message (BAM) for Vehicle ID longer than 8 Byte 68
2.2 Example SW Identification for buses and/or trucks 70
2.3 Example FMS Tell Tale Status 71
3 OVERVIEW MESSAGES 73

1 Parameters for FMS gateway (according to SAE J1939)

always MSB (Most Significant BIT) First

1.1 Parameters for Bus and Truck FMS-Standard

1.1.1 Fuel Consumption: LFC

0x00FEE9								PGN Hex
65,257								PGN
1000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
				8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	Bit No
Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Engine total fuel used 0,5 L / Bit gain 0 L offset SPN 250	Engine total fuel used 0,5 L / Bit gain 0 L offset SPN 250	Engine total fuel used 0,5 L / Bit gain 0 L offset SPN 250	Engine total fuel used 0,5 L / Bit gain 0 L offset SPN 250	Name values values values SPN

<u>Description acc. SAE J 1939:</u>

Total Fuel Used: Accumulated amount of fuel used during vehicle operation.

<u>Additional comment:</u>

Calculated values given as indications, not as contractual values.

Might be set to “not available” if the High Resolution Fuel Consumption is available

The mentioned resolution is not related to the accuracy of the signal

1.1.2 Dash Display 1: DD1

0x00FEFC								PGN Hex
65,276								PGN
1000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
	8 7 6 5 4 3 2 1					8 7 6 5 4 3 2 1		Bit No
Not used for (Bus) FMS-Standard	Fuel Level 1 0,4 % / Bit gain 0 % offset SPN 96	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Fuel Level 2 0,4 % / Bit gain 0 % offset SPN 38	Not used for (Bus) FMS-Standard	Name values values values SPN

Description acc. SAE J 1939:

Fuel Level: Ratio of volume of fuel to the total volume of fuel storage container.
When Fuel Level 2 (SPN 38) is not used, Fuel Level 1 (SPN 96) represents the total fuel in all fuel storage containers.
When Fuel Level 2 is used, Fuel Level 1 represents the fuel level in the primary or left-side fuel storage container.

Additional comment:

Fuel Level 2 is only used if different fuel types in different storage container

Fuel Level 1 represents always the fuel used for driving without indication of the location of the storage container.

Fuel Level 2 is not indicating the location of the storage container

The mentioned resolution is not related to the accuracy of the signal

1.1.3 Electronic Engine Controller #1: EEC1

0x00F004								PGN Hex
61,444								PGN
20 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
		8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1				Bit No
Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Actual Engine – Percent Torque 1 % / Bit -125 % offset SPN 513	Engine speed 0.125 rpm / Bit gain 0 rpm offset SPN 190	Engine speed 0.125 rpm / Bit gain 0 rpm offset SPN 190	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Name Name values values values SPN

Description acc. SAE J 1939:

Actual Engine – Percent Torque: The calculated output torque of the engine. The data is transmitted in indicated torque as a percent of reference engine torque (see the engine configuration message, PGN 65251).
The engine percent torque value will not be less than zero and it includes the torque developed in the cylinders required to overcome friction.

Engine Speed: Actual engine speed which is calculated over a minimum crankshaft angle of 720 degrees divided by the number of cylinders.

Additional comment:

The mentioned resolution is not related to the accuracy of the signal

1.1.4 Engine Hours, Revolutions: HOURS

0x00FEE5								PGN Hex
65,253								PGN
1000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1					Bit No
Engine total hours of Operation 0.05 h / Bit gain 0 h offset SPN 247	Engine total hours of Operation 0.05 h / Bit gain 0 h offset SPN 247	Engine total hours of Operation 0.05 h / Bit gain 0 h offset SPN 247	Engine total hours of Operation 0.05 h / Bit gain 0 h offset SPN 247	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Name values values values SPN

Description acc. SAE J 1939:

Engine total hours of Operation: Accumulated time of operation of engine.

Additional comment:

The mentioned resolution is not related to the accuracy of the signal

Counter is Engine running dependant

1.1.5 Vehicle Identification: VI

0x00FEEC								PGN Hex
65,260								PGN
10.000 ms								Rep. Rate
Variable 1-n	Variable 1-n	Variable 1-n	Variable 1-n	Variable 1-n	Variable 1-n	Variable 1-n	Variable 1-n	Byte No
								Bit No
Vehicle identification number ASCII up to 200 characters * = Delimiter SPN 237	Vehicle identification number ASCII up to 200 characters * = Delimiter SPN 237	Vehicle identification number ASCII up to 200 characters * = Delimiter SPN 237	Vehicle identification number ASCII up to 200 characters * = Delimiter SPN 237	Vehicle identification number ASCII up to 200 characters * = Delimiter SPN 237	Vehicle identification number ASCII up to 200 characters * = Delimiter SPN 237	Vehicle identification number ASCII up to 200 characters * = Delimiter SPN 237	Vehicle identification number ASCII up to 200 characters * = Delimiter SPN 237	Name values values values SPN

Description acc. SAE J 1939:
Vehicle identification number: Vehicle Identification Number (VIN) as assigned by the vehicle manufacturer. NOTE The ASCII character "*" is reserved as a delimiter.

Annotations:

1) If the Vehicle ID is up to 8 Bytes (including) then it is broadcasted with PGN 00FEEC containing the vehicle ID and filled with “FF” at the unused bytes.
2) If the Vehicle ID contains more than 8 Bytes then a TP.CM (PGN 00EC00) with a minimum of two TP.DT (PGN 00EB00) will be used.

Information might be a sub-set of the full VIN. ie: only last 8 bytes are sent
see example 2.1

Additional comment:

1.1.6 FMS-standard Interface Identity / Capabilities: FMS

0x00FDD1								PGN Hex
64,977								PGN
10.000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1				Bit No
Reserved for (Bus) FMS-Standard	SW-version supported Version number in the format ab.cd where this byte represents “a” ASCII SPN 2806	SW-version supported Version number in the format ab.cd where this byte represents “b” ASCII SPN 2806	SW-version supported Version number in the format ab.cd where this byte represents “c” ASCII SPN 2806	SW-version supported Version number in the format ab.cd where this byte represents “d” ASCII SPN 2806	Reserved for (Bus) FMS-Standard	Reserved for (Bus) FMS-Standard	Reserved for (Bus) FMS-Standard	Name values values values values values SPN
Requests supported 00b = request is not supported 01b= request is supported 10b = reserved 11b = don´t care SPN 2805								Name values values values values values values SPN
Diagnostics supported 00b = diagnostics is not supported 01b = diagnostics is supported 10b = reserved 11b = don´t care SPN 2804								Name values values values values values values SPN

Description acc. SAE J 1939:

Information which specifies the capabilities of the Fleet Management System (FMS) - standard interface device.
This PGN typically is sourced from the network interconnect FMS - standard interface device.
Requests supported: Status signal which indicates if the FMS Vehicle Interface (FMS Gateway) will respond to requests from the FMS device for the PGNs listed in the FMS Interface Specification.
This mode is to support FMS gateway devices that only operate in a 'Request' mode.
The FMS PGNs may also be broadcast periodically in this mode.
The FMS Gateway will NOT support the requests for information not included in the FMS Interface Specification onto the vehicle network..
Diagnostics supported: Status signal which indicates if the FMS Vehicle Interface (FMS Gateway) supports the handling of diagnostic messages from the vehicle network onto the FMS network.
The FMS gateway does NOT support the re-broadcast of diagnostics messages present on the vehicle network.
If this 'FMS-standard Diagnostics Supported' feature is supported by the FMS Gateway, the FMS Gateway will support the requests for diagnostics information (from the FMS device) onto the vehicle network and pass the responses onto the FMS network.
Note: This feature of the FMS Gateway is independent of the 'FMS-standard Requests Supported'. The FMS Gateway may support diagnostics without supporting the 'FMS-standard Requests Supported' function, or visa-versa..
FMS-standard SW-version supported: Information that identifies which issue level of the FMS-standard document the software included in the FMS gateway supports. Four bytes, representing ab.cd type revision level identification.
Version number in the format ab.cd where byte2 and 3 represent the version number for trucks “ab” (ASCII)
Byte 4 and 5 represent the version for buses and coaches “cd”(ASCII)
“00” represents “not supported”
For example, FMS-standard version 02.06 means the fms gateway supports version 02 of truck fms-standard and version 06 of bus fms-standard.

Additional comment:

See example 2.2

1.1.7 High Resolution Vehicle Distance: VDHR

0x00FEC1								PGN Hex
65,217								PGN
1000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1					Bit No
High resolution total vehicle distance 5 m / Bit gain 0 m offset SPN 917	High resolution total vehicle distance 5 m / Bit gain 0 m offset SPN 917	High resolution total vehicle distance 5 m / Bit gain 0 m offset SPN 917	High resolution total vehicle distance 5 m / Bit gain 0 m offset SPN 917	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Name values values values values SPN

Description acc. SAE J 1939:

High resolution total vehicle distance: Accumulated distance travelled by the vehicle during its operation.

Additional comment:

The mentioned resolution is not related to the accuracy of the signal

1.1.8 Tachograph : TCO1

0x00FE6C								PGN Hex
65,132								PGN
20 ms/ 50 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1			8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	Bit No.
Vehicle motion 00b = Vehicle motion not detected 01b = vehicle motion detected SPN 1611	Vehicle Overspeed 00b = No overspeed 01b = Overspeed SPN 1614	Not used for (Bus) FMS-Standard could be sent as "not available" or "don´t care"	Direction indicator 00b = Forward 01b = Reverse SPN 1619	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Tachogr. vehicle speed 1/256 km/h Bit gain 0 km/h offset SPN 1624	Tachogr. vehicle speed 1/256 km/h Bit gain 0 km/h offset SPN 1624	Name values values values values SPN
Driv. 2 working state 000b = Rest 001b =Driver available 010b = Work 011b = Drive 110b = Error 111b = not available SPN 1613	Driver 1 card 00b = Card not present 01b= Card present SPN 1615	Driver 2 card 00b = Card not present 01b= Card present SPN 1616	Tachograph performance 00b = Normal performance 01b = Performance analysis SPN 1620					Name values values values values values values values SPN
Driv. 1 working state 000b = Rest 001b =Driver available 010b = Work 011b = Drive 110b = Error 111b = not available SPN 1612	Driv. 1 time rel states 0000b = normal 0001b = 15 min bef. 4 ½ h 0010b = 4 ½ h reached 0011b = 15 min bef. 9 h 0100b = 9 h reached 0101b = 15 min bef. 16 h 0110b = 16h reached 1110b = Error 1111b = not available SPN 1617	Driv 2 time rel. states 0000b = normal 0001b = 15 min bef. 4 ½ h 0010b = 4 ½ h reached 0011b=15 min before 9 h 0100b = 9 h reached 0101b = 15 min bef. 16 h 0110b = 16h reached 1110b = Error 1111b = not available SPN 1618	Handling information 00b = no handling information 01b = handling information SPN 1621					Name values values values values values values values values SPN
			System event 00b = no tachogr. Event 01b = tachogr. Event SPN 1622					Name values values SPN

Description acc. SAE J 1939:

Vehicle motion: Indicates whether motion of the vehicle is detected or not.
Driver 2 Working State: State of work of the driver.
Driver 1 Working State: State of work of the driver.
Vehicle Overspeed: Indicates whether the vehicle is exceeding the legal speed limit set in the tachograph.
Driver 1 Card: Indicates the presence of a driver card.
Driver 1 Time Related Status: Indicates if the driver approaches or exceeds working time limits (or other limits).
Driver 2 Card: Indicates the presence of a driver card.
Driver 2 Time Related Status: Indicates if the driver approaches or exceeds working time limits (or other limits).
Direction Indicator: Indicates the direction of the vehicle.
Tachograph Performance: Indicates the tachograph performance; including electronic or mechanical analysis, instrument analysis, speed sensor analysis, mass storage analysis, and printer analysis.
Handling Information: Indicates that handling information is present. Information could include 'no printer paper', 'no driver card', etc.
System Event: Indicates that a tachograph event has occurred. This may include power supply interruption, interruption of the speed sensor, incorrect data on the driver card, driving without a driver card, illegal removal of a driver card, insertion of a driver card during driving, and time adjustment.
Tachograph Vehicle Speed: Speed of the vehicle registered by the tachograph.

Additional comment:

Tachograph vehicle speed might differ from the wheel based speed

The availability of the value direction indicator (SPN 1619) is tachograph dependant.
At the issuing date of this document the tachographs are not supporting this value.

Only available if a digital tachograph is present

1.1.9 Engine Temperature 1: ET1

0x00FEEE								PGN Hex
65,262								PGN
1000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 7 6 5 4 3 2 1								Bit No
Engine coolant temperature 1 °C / Bit gain - 40 °C offset SPN 110	Not used for (Bus) FMS-Standard	Not used for FMS-Standard	Not used for FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Name values values values SPN

Description acc. SAE J 1939:

Engine Coolant Temperature: Temperature of liquid found in engine cooling system.

Additional comment:

The mentioned resolution is not related to the accuracy of the signal

1.1.10 Ambient Conditions: AMB

0x00FEF5								PGN Hex
65,269								PGN
1000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
			8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1				Bit No
Not used for (Bus) FMS-Standard	Not used for FMS-Standard	Not used for FMS-Standard	Ambient Air Temperature 0.03125 °C / Bit gain - 273 °C offset SPN 171	Ambient Air Temperature 0.03125 °C / Bit gain - 273 °C offset SPN 171	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Name Name values values values values SPN

Description acc. SAE J 1939:

Ambient Air Temperature: Temperature of air surrounding vehicle.

Additional comment:

The mentioned resolution is not related to the accuracy of the signal.

1.1.11 Driver's Identification: DI

0x00FE6B								PGN Hex
65,131								PGN
10000 ms								Rep. Rate
Variable 1-n	Variable 1-n	Variable 1-n	Variable 1-n	Variable 1-n	Variable 1-n	Variable 1-n	Variable 1-n	Byte No
8-1	8 - 1	8 – 1	8-1	8 - 1	8 - 1	8 - 1	8 - 1	Bit No.
Driver 1 identification Driver 2 identification SPN 1625/1626	Driver 1 identification Driver 2 identification SPN 1625/1626	Driver 1 identification Driver 2 identification SPN 1625/1626	Driver 1 identification Driver 2 identification SPN 1625/1626	Driver 1 identification Driver 2 identification SPN 1625/1626	Driver 1 identification Driver 2 identification SPN 1625/1626	Driver 1 identification Driver 2 identification SPN 1625/1626	Driver 1 identification Driver 2 identification SPN 1625/1626	Name Name values values values values SPN

Description acc. SAE J 1939:

Field: a Driver 1 Identification Delimiter (ASCII '*') b Driver 2 Identification Delimiter (ASCII '*')
NOTE - If only driver card 1 is present, only the parameter driver 1 identification and two delimiters shall be transmitted.
If only driver card 2 is present, a delimiter followed by parameter driver 2 identification and the second delimiter shall be transmitted.
If no driver cards are present, only the two delimiters shall be sent."

Additional comment:

The driver ID is only available if a digital tachograph is present.
Driver ID = Issuing member state + CardNumber = 3 + 16 Byte (acc. ISO 16844)
If a driver ID is available the message is sent with a Broadcast Announce Message (BAM)
If no driver cards are present then it is broadcasted with PGN 00FE6B (8Byte) containing two delimiters and filled with “FF” at the unused bytes.

Difference to SAE: broadcast instead of on request

The information is sent by the tachograph. Depending on the tachograph brand the information is not immediately available after insertion of the driver card.

1.1.12 Fuel Economy: LFE

0x00FEF2								PGN Hex
65,266								PGN
100 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1					Bit No.
Fuel Rate 0.05 L/h per bit 0 offset 0 to 3,212.75 L/h SPN 183	Fuel Rate 0.05 L/h per bit 0 offset 0 to 3,212.75 L/h SPN 183	Instantaneous Fuel Economy 1/512 km/L per bit 0 offset 0 to 125,5 km/L SPN 184	Instantaneous Fuel Economy 1/512 km/L per bit 0 offset 0 to 125,5 km/L SPN 184	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Name Name values values values values values SPN

Description acc. SAE J 1939:

Fuel rate: Amount of fuel consumed by engine per unit of time

Instantaneous Fuel Economy: Current fuel economy at current vehicle velocity

Additional comment:

Calculated values given as indications, not as contractual values.

The mentioned resolution is not related to the accuracy of the signal

1.1.13 Air Supply Pressure : AIR1

0x00FEAE								PGN Hex
65,198								PGN
1000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
		8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1					Bit No.
Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Service Brake Air Pressure Circuit #1 8 kPa/Bit 0 offset SPN 1087	Service Brake Air Pressure Circuit #2 8 kPa/Bit 0 offset SPN 1088	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Name Name values values values values SPN

Description acc. SAE J 1939:

Service Brake Air Pressure Circuit #1: The pneumatic pressure in the service brake circuit or reservoir #1.
Service Brake Air Pressure Circuit #2: The pneumatic pressure in the service brake circuit or reservoir #2.

Additional comment:

The mentioned resolution is not related to the accuracy of the signal

1.1.14 High Resolution Fuel Consumption (Liquid): HRLFC

0x00FD09								PGN Hex
64,777								PGN
1000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
				Bit 8 - 1	Bit 8 - 1	Bit 8 - 1	Bit 8 - 1	Bit No.
Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	High resolution engine total fuel used 0.001 L/bit 0 offset 0 to 4,211,081.215 L SPN 5054	High resolution engine total fuel used 0.001 L/bit 0 offset 0 to 4,211,081.215 L SPN 5054	High resolution engine total fuel used 0.001 L/bit 0 offset 0 to 4,211,081.215 L SPN 5054	High resolution engine total fuel used 0.001 L/bit 0 offset 0 to 4,211,081.215 L SPN 5054	Name Name values values values values SPN

Description acc. SAE J 1939:

Engine fuel consumption accumulators

High resolution engine total fuel used: Accumulated amount of fuel used during vehicle operation. High resolution used for calculations and fleet management systems.

Additional comment:

Is implemented if technical possible.

Calculated values given as indications, not as contractual values.

The mentioned resolution is not related to the accuracy of the signal.

1.1.15 Aftertreatment 1 Diesel Exhaust Fluid Tank 1 Information: AT1T1I

0x00FE56								PGN Hex
65,110								PGN
1000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 7 6 5 4 3 2 1								Bit No
Aftertreatment 1 Diesel Exhaust Fluid Tank 1 Level 0 % = Empty 100% = Full 0.4 % / bit, 0 offset SPN 1761	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Name values values values values values SPN

Description acc. SAE J 1939:

Ratio of volume of diesel exhaust fluid to the total volume of diesel exhaust fluid storage container.

Additional comment:

The mentioned resolution is not related to the accuracy of the signal.

1.1.16 FMS Tell Tale Status: FMS1

0x00FD7D								PGN Hex
64,893								PGN
1000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	Bit No.
Telltale Block ID see table for Block ID and Telltale ID 1111b = don't care	Telltale Status 2 000b = off 001b = Cond. Red 010b = Cond. Yellow 011b = Cond. Info 100b–110b = Reserved 111b = not available	Telltale Status 4 000b = off 001b = Cond. Red 010b = Cond. Yellow 011b = Cond. Info 100b–110b = Reserved 111b = not available	Telltale Status 6 000b = off 001b = Cond. Red 010b = Cond. Yellow 011b = Cond. Info 100b–110b = Reserved 111b = not available	Telltale Status 8 000b = off 001b = Cond. Red 010b = Cond. Yellow 011b = Cond. Info 100b–110b = Reserved 111b = not available	Telltale Status 10 000b = off 001b = Cond. Red 010b = Cond. Yellow 011b = Cond. Info 100b–110b = Reserved 111b = not available	Telltale Status 12 000b = off 001b = Cond. Red 010b = Cond. Yellow 011b = Cond. Info 100b–110b = Reserved 111b = not available	Telltale Status 14 000b = off 001b = Cond. Red 010b = Cond. Yellow 011b = Cond. Info 100b–110b = Reserved 111 = not available	Name values values values values values values
Telltale Status 1 000b = off 001b = Cond. Red 010b = Cond. Yellow 011b = Cond. Info 100b–110b = Reserved 111b = not available	Telltale Status 3 000b = off 001b = Cond. Red 010b = Cond. Yellow 011b = Cond. Info 100b–110b = Reserved 111b = not available	Telltale Status 5 000b = off 001b = Cond. Red 010b = Cond. Yellow 011b = Cond. Info 100b–110b = Reserved 111b = not available	Telltale Status 7 000b = off 001b = Cond. Red 010b = Cond. Yellow 011b = Cond. Info 100b–110b = Reserved 111b = not available	Telltale Status 9 000b = off 001b = Cond. Red 010b = Cond. Yellow 011b = Cond. Info 100b–110b = Reserved 111b = not available	Telltale Status 11 000b = off 001b = Cond. Red 010b = Cond. Yellow 011b = Cond. Info 100b–110b = Reserved 111b = not available	Telltale Status 13 000b = off 001b = Cond. Red 010b = Cond. Yellow 011b = Cond. Info 100b–110b = Reserved 111b = not available	Telltale Status 15 000b = off 001b = Cond. Red 010b = Cond. Yellow 011b = Cond. Info 100b–110b = Reserved 111b = not available	Name values values values values values values
Not defined (set to "1")	Not defined (set to "1")	Not defined (set to "1")	Not defined (set to "1")	Not defined (set to "1")	Not defined (set to "1")	Not defined (set to "1")	Not defined (set to "1")	Name values values values values values

Table for Telltale status:

Block ID	Telltale Status	Telltale ID	ISO No.	Name	Mandatory Truck only		Block ID	Telltale Status	Telltale ID	ISO No.	Name	Mandatory Truck only
0	1	1	27	Cooling air conditioning			2	1	31	2441	Steering failure	
0	2	2	82	High beam, main beam			2	2	32	2461	Height Control (Levelling)	
0	3	3	83	Low beam, dipped beam			2	3	33	2574	Retarder	
0	4	4	84	Turn signals			2	4	34	2596	Engine Emission system failure (Mil indicator)	
0	5	5	85	Hazard warning			2	5	35	2630	ESC indication	
0	6	6	100	Provision for the disabled or handicapped persons			2	6	36	no	Brake lights	
0	7	7	238	Parking Brake	X		2	7	37	no	Articulation	
0	8	8	239	Brake failure/brake system malfunction			2	8	38	no	Stop Request	
0	9	9	242	Hatch open			2	9	39	no	Pram request	
0	10	10	245	Fuel level	X		2	10	40	no	Bus stop brake	
0	11	11	246	Engine coolant temperature	X		2	11	41	2946	AdBlue level	
0	12	12	247	Battery charging condition			2	12	42	no	Raising	
0	13	13	248	Engine oil	X		2	13	43	no	Lowering	
0	14	14	456	Position lights,side lights			2	14	44	no	Kneeling	
0	15	15	633	Front fog light			2	15	45	no	Engine compartment temperature	
1	1	16	634	Rear fog light			3	1	46	no	Auxillary air pressure	
1	2	17	637	Park Heating			3	2	47	2432	Air filter clogged	
1	3	18	640	Engine / Mil indicator	X		3	3	48	2452	Fuel filter differential pressure	
1	4	19	717	Service, call for maintenance			3	4	49	249	Seat belt	
1	5	20	1168	Transmission fluid temperature			3	5	50	no	EBS	
1	6	21	1396	Transmission failure/malfunction			3	6	51	2682	Lane departure indication	
1	7	22	1407	Anti-lock brake system failure			3	7	52	no	Advanced emergency braking system	
1	8	23	1408	Worn brake linings			3	8	53	2581	ACC	
1	9	24	1422	Windscreen washer fluid/windshield washer fluid			3	9	54	no	Trailer connected	
1	10	25	1434	Tire failure/malfunction			3	10	55	2444/2445	ABS Trailer 1,2	
1	11	26	1603	Malfunction/general failure			3	11	56	2108	Airbag	
1	12	27	2426	Engine oil temperature			3	12	57	no	EBS Trailer 1,2	
1	13	28	2427	Engine oil level			3	13	58	no	Tachograph indication	
1	14	29	2429	Engine coolant level			3	14	59	2649	ESC switched off	
1	15	30	2440	Steering fluid level			3	15	60	no	Lane departure warning switched off	

Block ID	Telltale Status	Telltale ID	ISO No.	Name	Mandatory Truck only		Block ID	Telltale Status	Telltale ID	ISO No.	Name	Mandatory Truck only
4	1	61	2433	Engine emission filter (Soot Filter)								
4	2	62	2633	Electric motor failures								
4	3	63	no	AdBlue tampering								
4	4	64	no	Multiplex System								
4	5	65	2632	Battery pack								
4	6	66	6042	High voltage system caution								
4	7	67	3129	Battery pack temperature								
4	8	68	2639	Limited performance electric motor								
4	9	69	2455	Battery pack cooling								
4	10	70	no	Charging status								
4	11	71	no	Traction battery balancing								
4	12	72	no	Pantograph status								
4	13	73	no	Fire detection / alarm								
4	14	74	no	Recuperation disturbed								
4	15	75	no	High Voltage Battery Thermal Event								

Annotations:

Tell Tale Status:
The Tell Tale Status information is derived from information displayed to the driver's dashboard.
The tell tale number is related to the symbols in the ISO 7000 document and not to the description.
The not related to ISO 7000 is stated with "no"
There are three possible conditions stated: Red ("R"), Yellow ("Y"), Info ("I"). The interpretation of the status is manufacturer dependant and might be different.
For details please refer to the manufacturers' document.
The symbols used in the dash display of each manufacturer might vary from ISO symbols.
The lamp characteristic (e.g., flashing) is not reflected in the tell tale information.
Some information might be not shown to the driver as a tell-tale.
The status information is present as long the status is valid.

Additional comment:
Due to the repetition rate of the message, it is not guaranteed to include all intermittent signals.*

The message is mandatory for block 0-1 (for trucks) however some tell tale status information are not mandatory
Further blocks are optional
"skipping blocks" (e.g., if all tell tales of a block are not available) is allowed (e.g. sending block 0,1,4)

Charging status: The information is from the vehicle point of view and might be different from the SPN 13171 which is the view from the charging station
Traction battery balancing: The balancing is mainly done automatically or initiated from the battery system and might be needed to be connected to a charging station. Depending on the OEM of the vehicle the balancing might be always on (as well during driving) and handled by the vehicle system.
Pantograph status: indication if the pantograph has a fault/warning
Fire detection / alarm: Extinguisher is not included, Yellow state could be used for indicating that there is a problem with the fire detection system
Recuperation disturbed: Indicate that recuperation is disabled/disturbed/limited in BEVs.
High Voltage Battery Thermal Event: Information about a possible thermal issue detection.

see example in 2.3.

1.1.17 Electronic Brake Controller 1: EBC1

0x00F001								PGN Hex
61,441								PGN
100 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
	8 7 6 5 4 3 2 1							Bit No.
Not used for (Bus) FMS-Standard	Brake Pedal Position 0,4 % / Bit gain 0 % offset SPN 521	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Name Name values values values values SPN

Description acc. SAE J 1939:

Ratio of brake pedal position to maximum pedal position.
Used for electric brake applications.
0% means no braking.
Also when there are two brake pedals on the machine (Left Brake Pedal Position SPN 3033 and Right Brake Pedal Position SPN 3032) the maximum of the two should be transmitted for Brake Pedal Position.

Additional comment:

The mentioned resolution is not related to the accuracy of the signal.

1.1.18 Electronic Engine Controller 14: EEC14

0x00FDC2								PGN Hex
64,962								PGN
10000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
						8 7 6 5 4 3 2 1		Bit No.
Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Fuel Type 256 states/8 bit SPN 5837	Not used for (Bus) FMS-Standard	Name Name values values SPN

Description acc. SAE J 1939:

Type of fuel currently being utilized by the vehicle.

See table below.

Data Byte	Scaling	External Display	Data Byte	Scaling	External Display
0x00h	Not available	NONE	0x12h	Hybrid vehicle using gasoline engine on ethanol	HYB_ETH
0x01h	Gasoline/petrol	GAS	0x13h	Hybrid vehicle using diesel engine	HYB_DSL
0x02h	Methanol	METH	0x14h	Hybrid vehicle using battery	HYB_ELEC
0x03h	Ethanol	ETH	0x15h	Hybrid vehicle using battery and combustion engine	HYB_MIX
0x04h	Diesel	DSL	0x16h	Hybrid vehicle in regeneration mode	HYB_REG
0x05h	Liquefied Petroleum Gas (LPG)	LPG	0x17h	Natural Gas (NG)	NG
0x06h	Compressed Natural Gas (CNG)	CNG	0x18h	Bi-fuel vehicle using NG	BI_NG
0x07h	Propane	PROP	0x19h	Bi-fuel using diesel	
0x08h	Battery/electric	ELEC	0x1Ah	Natural Gas (Compressed or Liquefied Natural Gas)	
0x09h	Bi-fuel vehicle using gasoline	BI_GAS	0x1Bh	Dual Fuel – Diesel and CNG	
0x0Ah	Bi-fuel vehicle using methanol	BI_METH	0x1Ch	Dual Fuel – Diesel and LNG	
0x0Bh	Bi-fuel vehicle using ethanol	BI_ETH	0x1Dh	Fuel Cell Utilizing Hydrogen	
0x0Ch	Bi-fuel vehicle using LPG	BI_LPG	0x1Eh	Hydrogen Internal Combustion Engine	
0x0Dh	Bi-fuel vehicle using CNG	BI_CNG	0x1Fh	Kerosene	
0x0Eh	Bi-fuel vehicle using propane	BI_PROP	0x20h	Heavy Fuel Oil	
0x0Fh	Bi-fuel vehicle using battery	BI_ELEC	0x21h- 0xFDh	ISO/SAE reserved	
0x10h	Bi-fuel vehicle using battery and combustion engine	BI_MIX	0xFEh	Error	
0x11h	Hybrid vehicle using gasoline engine	HYB_GAS	0xFFh	Not available (Parameter not supported)	

EXAMPLE: If a Bi-fuel vehicle has less than 10% ethanol in the gasoline/petrol, then the external test equipment shall display state 0x09. In such a case, the system would be using gasoline/petrol (fuel) tables and OBD thresholds. SPN 5458 should either artificially display 0% or some number 10% or less if the system has such resolution. If the ethanol in the gasoline/petrol is above 10%, then the external test equipment shall display state 0x0B. SPN 5458 shall report the calculated ethanol/alcohol percentage.

Note: "Not available" (Data 0x00) shall be used for the following conditions where no fuel is being utilized:

- Ignition Key On/Engine Off, for example, for bi-fuel vehicles when the vehicle cannot identify which fuel is being used before engine start
- Engine stopped for vehicle using Start/Stop strategy or for hybrid vehicle with engine stopped If FUEL_TYP is utilized on a single fuel vehicle, it can continue to indicate the previously utilized fuel type while the engine is in Decel Fuel Cut Off (DFCO) mode. It is not required to indicate 0x00 while in DFCO.

Regarding fuel type 0x17 - Natural Gas - this applies to any type of natural gaseous fuel when the source (CNG, LPG, or other) is not relevant.

Regarding fuel type 0x18 - Bi-fuel vehicle using NG; this applies to bi-fuel vehicles running Diesel and Natural Gas, where the source of the natural gaseous fuel is not relevant.

Regarding fuel type 20h - Heavy Fuel Oil is a Maritime fuel classification for pure or nearly pure residual oil, roughly equivalent to No. 6 fuel oil Marine diesel oil contains some heavy fuel oil, unlike regular diesels.

1.1.19 Fuel Consumption (Gaseous): GFC

0x00FEAF								PGN Hex
65,199								PGN
1000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
				Bit 8-1	Bit 8-1	Bit 8-1	Bit 8-1	Bit No.
Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Total Fuel Used (Gaseous) 0.5 kg/bit 0 to 2,105,540,607.5 kg SPN 1040	Total Fuel Used (Gaseous) 0.5 kg/bit 0 to 2,105,540,607.5 kg SPN 1040	Total Fuel Used (Gaseous) 0.5 kg/bit 0 to 2,105,540,607.5 kg SPN 1040	Total Fuel Used (Gaseous) 0.5 kg/bit 0 to 2,105,540,607.5 kg SPN 1040	Name Name values values values values SPN

Description acc. SAE J 1939:

Total fuel consumed (trip drive fuel + trip PTO governor moving fuel + trip PTO governor non-moving fuel + trip idle fuel) over the life of the engine.

Additional comment:

The mentioned resolution is not related to the accuracy of the signal

1.1.20 Electronic Retarder Controller 1: ERC1

0x00F000								PGN Hex
61,440								PGN
100 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1					8 7 6 5 4 3 2 1		Bit No
Retarder Torque Mode 16 states/4 bit, 0 offset SPN 900	Actual Retarder - Percent Torque 1 %/bit, -125 % offset SPN 520	Not used for FMS-Standard	Not used for FMS-Standard	Not used for FMS-Standard	Not used for FMS-Standard	Retarder Selection, non-engine 0.4 %/bit, 0 % offset SPN 1716	Not used for FMS-Standard	Name values values values SPN
Not used for FMS-Standard								

Description acc. SAE J 1939:

Retarder Torque Mode: State signal which indicates which retarder torque mode is currently generating, limiting, or controlling the torque.
Note that the modes are not in prioritized order. Not all modes may be relevant for a given device. Some devices may not implement all functions.
Mode 0000b means “No request”: retarder torque = 0 (no braking). See Appendix D in SAE documentation
Modes 0001b to 1110b indicate that there is either a torque request or the identified function is currently controlling the retarder:
retarder torque may range from 0 (no braking) to the upper limit.

Actual Retarder - Percent Torque: Actual braking torque of the retarder as a percent of retarder configuration reference torque SPN 556.

Retarder Selection, non-engine: The “Retarder Selection, non-engine” is the position of the driver’s selector for retarders that are not part of the engine system, expressed as percent and determined by the ratio of current position to the maximum possible position. The physical device may be a lever, rotary dial, combination of switches, or other device that the driver can use to select the type or amount of retardation needed.

Additional comment:

For SPN 1716: The value is related to the driver’s selection of a retarder (engine and/or drive line).

1.1.21 Vehicle Weight: VW

0x00FEEA								PGN Hex
65,258								PGN
1000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1						Bit No
Axle location Bit-mapped position number counting front to back facing forward Fh = not available SPN 928	Axle weight 0.5 kg / Bit gain 0 kg offset SPN 582	Axle weight 0.5 kg / Bit gain 0 kg offset SPN 582	Not used for FMS-Standard	Not used for FMS-Standard	Not used for FMS-Standard	Not used for FMS-Standard	Not used for FMS-Standard	Name values values values values values values SPN
Tire location Bit-mapped counting left to right facing forward Fh = not available SPN 928								Name values values values values values values SPN

Description acc. SAE J 1939:

Axle / Tire Location: To identify to which of several similar devices (such as tires or fuel tanks) the information applies.
The low order 4 bits represent a position number, counting left to right when facing in the direction of normal vehicle travel (forward).
The high order 4 bits represent a position number, counting front to back on the vehicle.
The value 0xFF indicates not available.
It is recommended that output devices add 1 to the position number (range 1 to 15, not 0 to 14) for use by drivers and service technicians.
Examples: Tire pressure for location 0x00 would be left front tire.
Tire pressure for location 0x23 would be right outside rear rear on a 3-axle tractor with dual axle per side (3rd axle, 4th tire)
Axel weight: Total mass imposed by the tires on the road surface at the specified axle.

1.1.22 EVSE 1 DC Status 1 : EVSE1DCS1

0x00FAB8								PGN Hex
64,184								PGN
1000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 7 6 5 4 3 2 1								Bit No.
DC Charging State 0 = Idle. No EVSE connected. 1 = Charging. 2 = Standby. 3-9 = reserved 10 = Initialising 11 = Pre-charge 12 = Terminating 13 = Battery failure. An error condition due to the battery state (high temperature, etc.) 14 = EVSE fault. An error condition due to the EVSE. 15 = Not available. SPN 13171	Not used in FMS-Standard	Not used in FMS-Standard	Not used in FMS-Standard	Not used in FMS-Standard	Not used in FMS-Standard	Not used in FMS-Standard	Not used in FMS-Standard	Name values values values values values values values values values values values values values SPN

Description acc. SAE J 1939:

DC Charging State: Overall State of the DC charging system.

Additional Comment:

Please note that the information is reported by the EVSE (Electric Vehicle Supply Equipment) beside status 0 and 15.
The mentioned resolution is not related to the accuracy of the signal.

1.1.23 EV Operator Indicators 1 : EVOI1

0x00FC69								PGN Hex
64,617								PGN
Every 10 s and on change but no faster than 1 s								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1							Bit No.
HVESS Estimated Remaining Distance 0.25 km/bit 0 offset SPN 15268	HVESS Estimated Remaining Distance 0.25 km/bit 0 offset SPN 15268	Not used in FMS-Standard	Not used in FMS-Standard	Not used in FMS-Standard	Not used in FMS-Standard	Not used in FMS-Standard	Not used in FMS-Standard	Name Name values values SPN

Description acc. SAE J 1939:

HVESS Estimated Remaining Distance: Estimated remaining distance based on existing available HVES system capacity.

Additional Comment:

The mentioned resolution is not related to the accuracy of the signal.

1.1.24 Tire Condition Message 1 : TIRE1

0x00FEF4								PGN Hex
65,268								PGN
10.000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1							Bit No.
Tire Location 256 states/8 bit 0 offset SPN 929	Tire Pressure 4 kPa/bit 0 offset SPN 241	Not used in FMS-Standard	Not used in FMS-Standard	Not used in FMS-Standard	Not used in FMS-Standard	Not used in FMS-Standard	Not used in FMS-Standard	Name values values SPN

Description acc. SAE J 1939:

Tire Location: Identifies which tire is associated with the parametric data in this PG.

The low order 4 bits represent a position number, counting left to right when facing in the direction of normal vehicle travel (forward).

The high order 4 bits represent a position number, counting front to back on the vehicle.

The value FFh indicates not available.

It is recommended that output devices add 1 to the position number (range 1 to 15, not 0 to 14) for use by drivers and service technicians.

Examples: Tire pressure for location 00h would be left front tire.

Tire pressure for location 23h would be right outside rear rear tire on a 3-axle tractor with dual axle per side (3rd axle, 4th tire).

Tire Pressure: Pressure at which air is contained in cavity formed by tire and rim.

Additional Comment:

The mentioned resolution is not related to the accuracy of the signal.

1.1.25 Operator Indicators : OI

0x00FEFF								PGN Hex
65,279								PGN
10 s or on change but no more often than 1 s								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 7 6 5 4 3 2 1		8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1					Bit No.
Water In Fuel Indicator 1 00b = No 01b = Yes 10b = Error 11b = Not available SPN 97	Not used in FMS-Standard	Fuel Supply Estimated Remaining Distance 1 km/bit 0 offset SPN 8428	Fuel Supply Estimated Remaining Distance 1 km/bit 0 offset SPN 8428	Not used in FMS-Standard	Not used in FMS-Standard	Not used in FMS-Standard	Not used in FMS-Standard	Name Name values values values values SPN

Description acc. SAE J 1939:

Fuel Supply Estimated Remaining Distance: Estimated remaining distance based on existing fuel supply.

Water In Fuel Indicator 1: Signal which indicates the presence of water in the fuel.

Additional Comment:

The mentioned resolution is not related to the accuracy of the signal.

The SPN 15268 is used for BEV instead of SPN 8428.

1.2 Parameters for Truck FMS-Standard

1.2.1 Cruise Control/Vehicle Speed 1: CCVS1

0x00FEF1								PGN Hex
65,265								PGN
100 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1			8 7 6 5 4 3 2 1		Bit No
Not used for FMS-Standard	Wheel based speed 1/256 km/h Bit gain 0 km/h offset SPN 84	Wheel based speed 1/256 km/h Bit gain 0 km/h offset SPN 84	Clutch switch 00b = pedal released 01b = pedal depressed SPN 598	Not used for FMS-Standard	Not used for FMS-Standard	Cruise Control States 000b - Off/Disabled 001b – Hold 010b – Accelerate 011b – Decelerate 100b – Resume 101b – Set 110b – Accel. Override 111b - Not available SPN 527	Not used for FMS-Standard	Name values values values values values values values values SPN
Parking Brake Switch 00b = Parking brake not set 01b = Parking brake set SPN 70			Brake switch 00b = pedal released 01b = pedal depressed SPN 597			PTO state 00000b = off/disabled 00101b = Set 11111b = not available SPN 976		Name values values values SPN
			Not used for FMS Standard					Name
			Cruise control active 00b = switched off 01b = switched on SPN 595					Name values values values SPN

Description acc. SAE J 1939:

Parking Brake Switch: Switch signal which indicates when the parking brake is set. In general the switch actuated by the operator's park brake control, whether a pedal, lever or other control mechanism.
Wheel Based Speed: Speed of the vehicle as calculated from wheel or tailshaft speed.
Clutch Switch: Switch signal which indicates that the clutch pedal is being pressed. It is necessary for a safe drivetrain behaviour that the clutch switch is set before the clutch is opened (cruise control function).
Brake Switch: Switch signal which indicates that the driver operated brake foot pedal is being pressed. This brake foot pedal is controlling the vehicles' service brake (total vehicle braking application, not park brakes). It is necessary for safe drivetrain behaviour that the switch activates before the physical braking components are activated (i.e. Disengage the cruise control function prior to the activation of friction brakes).
Cruise Control Active: Cruise control is switched on. It is not ensured that the engine is controlled by cruise control, as in the case of a large driver's demand the engine is controlled by the driver while cruise control is active (maximum selection of cruise control and driver's demand). The cruise control is set to 0 if a switch off condition occurs.
PTO state: This parameter is used to indicate the current state or mode of operation by the power takeoff (PTO) device. It needs to be ensured that each achieved state information be set up to be conveyed in at least one datalink message before a transition to another state is allowed.
Off/Disabled 00000b — Used to indicate that the PTO enable switch is in the off position.
Set 00101b — Used to indicate that the PTO device is establishing current speed as the operating speed (captured value).

Additional comment:

The cruise control conditions might vary on different brands.

Wheel based speed might vary from tacho speed.

The PTO state might be different over the brands (not comparable) due to different internal topology

Either SPN 3948 (PTO DE) or SPN 976 (CCVS) is sent. PTO DE message is preferred

The clutch switch information is depending on the gear box type *

The mentioned resolution is not related to the accuracy of the signal

1.2.2 Electronic Engine Controller #2: EEC2

0x00F003								PGN Hex
61,443								PGN
50 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1						Bit No
Not used for FMS-Standard	Accelerator pedal position 1 0,4 % / Bit gain 0 % offset SPN 91	Engine Percent Load At Current Speed 1 % / bit, 0 offset 0 to 125 % op. range SPN 92	Not used for FMS-Standard	Not used for FMS-Standard	Not used for FMS-Standard	Not used for FMS-Standard	Not used for FMS-Standard	Name values values values SPN

Description acc. SAE J 1939:

Accelerator Pedal Position: The ratio of actual position of the analogue engine speed/torque request input device (such as an accelerator pedal or throttle lever) to the maximum position of the input device. This parameter is intended for the primary accelerator control in an application. If an application has only one accelerator control, use SPN 91. For on-highway vehicles, this will typically be the operator's accelerator pedal. Although it is used as an input to determine powertrain demand, it also provides anticipatory information to transmission and ASR algorithms about driver actions.

Engine Percent Load At Current Speed
At Current Speed
The ratio of actual engine percent torque (indicated) to maximum indicated torque available at the current engine speed, clipped to zero torque during engine braking.

Additional comment:

The mentioned resolution is not related to the accuracy of the signal

1.2.3 Service Information: SERV

0x00FEC0								PGN Hex
65,216								PGN
1000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1						Bit No
Not used for FMS-Standard	Service distance 5 km / Bit gain -160 635 km offset SPN 914	Service distance 5 km / Bit gain -160 635 km offset SPN 914	Not used for FMS-Standard	Not used for FMS-Standard	Not used for FMS-Standard	Not used for FMS-Standard	Not used for FMS-Standard	Name values values values SPN

Description acc. SAE J 1939:

Service distance: The distance which can be travelled by the vehicle before the next service inspection is required. A negative distance is transmitted if the service inspection has been passed. The component that requires service is identified by the service component identification (see SPN 911-913, 1379, and 1584)

Additional comment:

The mentioned resolution is not related to the accuracy of the signal

1.2.4 PTO Drive Engagement: PTODE

0x00FDA4								PGN Hex
64,932								PGN
100 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
						8 7 6 5 4 3 2 1		Bit No.
Not used for FMS-Standard	Not used for FMS-Standard	Not used for FMS-Standard	Not used for FMS-Standard	Not used for FMS-Standard	Not used for FMS-Standard	At least one PTO engaged 00b No PTO drive is engaged 01b At least one PTO drive is engaged 10b Error 11b Not available SPN 3948	Not used for FMS-Standard	Name Name values values values values values values values values SPN
						Not used for FMS-Standard		

Description acc. SAE J 1939:

Information relating to the request for engagement, consent for engagement, and status of engagement of various specific physical PTO drives.

This message may be broadcast by one or all controllers involved in the enabling of a given PTO drive

At least one PTO engaged: Indicates that at least one PTO is engaged

Note: This parameter should only be sent by the controller that has knowledge of all PTO drives on the vehicle (e.g, the FMS gateway).

Individual PTO drive controllers should broadcast this parameter as "not available".

Additional comment:

Either SPN 3948 (PTO DE) or SPN 976 (CCVS) is sent. PTO DE message is preferred

1.2.5 Combination Vehicle Weight: CVW

0x00FE70								PGN Hex
65,136								PGN
10 000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
		8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1					Bit No.
Not used for FMS-Standard	Not used for FMS-Standard	Gross Combination Vehicle Weight 10 kg/bit 0 offset 0 to 642,550 kg SPN 1760	Gross Combination Vehicle Weight 10 kg/bit 0 offset 0 to 642,550 kg SPN 1760	Not used for FMS-Standard	Not used for FMS-Standard	Not used for FMS-Standard	Not used for FMS-Standard	Name Name values values values values values SPN

Description acc. SAE J 1939:

The total weight of the truck and all attached trailers.

Additional comment:

The mentioned resolution is not related to the accuracy of the signal

1.3 Parameters for Bus FMS-Standard

1.3.1 Cruise Control/Vehicle Speed: CCVS

0x00FEF1								PGN Hex
65,265								PGN
100 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1			8 7 6 5 4 3 2 1		Bit No
Not used in Bus FMS-Standard	Wheel based speed 1/256 km/h Bit gain 0 km/h offset SPN 84	Wheel based speed 1/256 km/h Bit gain 0 km/h offset SPN 84	Clutch switch 00b = pedal released 01b = pedal pressed SPN 598	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Cruise Control States 000b - Off/Disabled 001b – Hold 010b – Accelerate 011b – Decelerate 100b – Resume 101b – Set 110b – Accel. Override 111b - Not available SPN 527	Not used in Bus FMS-Standard	Name values values values values values values values values SPN
Parking Brake Switch 00b = Parking brake not set 01b = Parking brake set SPN 70			Brake switch 00b = pedal released 01b = pedal depressed SPN 597			Not used in Bus FMS-Standard		Name values values values values values values SPN
			Not used in Bus FMS-Standard					
			Cruise control active 00b = switched off 01b = switched on SPN 595					Name values values values values SPN

Description acc. SAE J 1939:

Parking Brake Switch: Switch signal which indicates when the parking brake is set. In general the switch actuated by the operator's park brake control, whether a pedal, lever or other control mechanism.
Wheel Based Speed: Speed of the vehicle as calculated from wheel or tailshaft speed.
Clutch Switch: Switch signal which indicates that the clutch pedal is being pressed. It is necessary for a safe drivetrain behaviour that the clutch switch is set before the clutch is opened (cruise control function).
Brake Switch: Switch signal which indicates that the driver operated brake foot pedal is being pressed. This brake foot pedal is controlling the vehicles' service brake (total vehicle braking application, not park brakes). It is necessary for safe drivetrain behaviour that the switch activates before the physical braking components are activated (i.e. Disengage the cruise control function prior to the activation of friction brakes).
Cruise Control Active: Cruise control is switched on. It is not ensured that the engine is controlled by cruise control, as in the case of a large driver's demand the engine is controlled by the driver while cruise control is active (maximum selection of cruise control and driver's demand). The cruise control is set to 0 if a switch off condition occurs.

Additional comment:

The cruise control conditions might vary on different brands.

Wheel based speed might vary from tacho speed.

The clutch switch information is depending on the gear box type *

The mentioned resolution is not related to the accuracy of the signal

1.3.2 Electronic Engine Controller #2 : EEC2

0x00F003								PGN Hex
61,443								PGN
50 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
	8 7 6 5 4 3 2 1							Bit No
Not used in Bus FMS-Standard	Accelerator pedal position 0,4 % / Bit gain 0 % offset SPN 91	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Name Name values values values values SPN

Description acc. SAE J 1939:

Accelerator Pedal Position: The ratio of actual position of the analogue engine speed/torque request input device (such as an accelerator pedal or throttle lever) to the maximum position of the input device. This parameter is intended for the primary accelerator control in an application. If an application has only one accelerator control, use SPN 91. For on-highway vehicles, this will typically be the operator's accelerator pedal. Although it is used as an input to determine powertrain demand, it also provides anticipatory information to transmission and ASR algorithms about driver actions.

Additional comment:

The mentioned resolution is not related to the accuracy of the signal.

1.3.3 Door Control 1: DC1

0x00FE4E								PGN Hex
65,102								PGN
100 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 7 6 5 4 3 2 1								Bit No.
Status 2 of doors 00b = all bus doors disabled 01b = at least 1 bus door enabled 10b = error 11b = not available SPN 3411	Not used in Bus-FMS-Standard	Not used in Bus-FMS-Standard	Not used in Bus-FMS-Standard	Not used in Bus-FMS-Standard	Not used in Bus-FMS-Standard	Not used in Bus-FMS-Standard	Not used in Bus-FMS-Standard	Name values values values values values values values SPN
Ramp/Wheel chairlift 00b = inside bus 01b = outside bus 10b = Error 11b = not available SPN 1820								Name values values values values values SPN
Position of doors 0000b = at least 1 door is open 0001b = closing last door 0010b= all doors closed 0011b-1101b not defined 1110b = Error 1111b = not available SPN 1821								Name values values values values values values values SPN

Description acc. SAE J 1939:

Status 2 of doors: Composite indication of all bus door statuses. Enabled means the bus doors are able to be automatically opened or closed..
Ramp/Wheel Chair Lift Position: Signal which indicates the actual position of the ramp / wheel chair lift.
Position of Doors: Signal which indicates the actual position of the doors.
Additional comment:
For the door configuration please contact the manufacturer of the vehicle.

1.3.4 Door Control 2: DC2

0x00FDA5								PGN Hex
64,933								PGN
100 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	Bit No.
Lock Status Door 2 00b = Unlocked 01b = Locked 10b = Error 11b = Not available SPN 3415	Open Status Door 3 00b = Closed 01b = Open 10b = Error 11b = Not available SPN 3419	Enable Status Door 4 00b = Disabled 01b = Enabled 10b = Error 11b = Not available SPN 3423	Lock Status Door 6 00b = Unlocked 01b = Locked 10b = Error 11b = Not available SPN 3427	Open Status Door 7 00b = Closed 01b = Open 10b = Error 11b = Not available SPN 3431	Enable Status Door 8 00b = Disabled 01b = Enabled 10b = Error 11b = Not available SPN 3435	Lock Status Door 10 00b = Unlocked 01b = Locked 10b = Error 11b = Not available SPN 3439	Not defined	Name values values values values values SPN
Enable Status Door 1 00b = Disabled 01b = Enabled 10b = Error 11b = Not available SPN 3414	Lock Status Door 3 00b = Unlocked 01b = Locked 10b = Error 11b = Not available SPN 3418	Open Status Door 4 00b = Closed 01b = Open 10b = Error 11b = Not available SPN 3422	Enable Status Door 5 00b = Disabled 01b = Enabled 10b = Error 11b = Not available SPN 3426	Lock Status Door 7 00b = Unlocked 01b = Locked 10b = Error 11b = Not available SPN 3430	Open Status Door 8 00b = Closed 01b = Open 10b = Error 11b = Not available SPN 3434	Enable Status Door 9 00b = Disabled 01b = Enabled 10b = Error 11b = Not available SPN 3438	Not defined	Name values values values values values SPN
Open Status Door 1 00b = Closed 01b = Open 10b = Error 11b = Not available SPN 3413	Enable Status Door 2 00b = Disabled 01b = Enabled 10b = Error 11b = Not available SPN 3417	Lock Status Door 4 00b = Unlocked 01b = Locked 10b = Error 11b = Not available SPN 3421	Open Status Door 5 00b = Closed 01b = Open 10b = Error 11b = Not available SPN 3425	Enable Status Door 6 00b = Disabled 01b = Enabled 10b = Error 11b = Not available SPN 3429	Lock Status Door 8 00b = Unlocked 01b = Locked 10b = Error 11b = Not available SPN 3433	Open Status Door 9 00b = Closed 01b = Open 10b = Error 11b = Not available SPN 3437	Enable Status Door 10 00b = Disabled 01b = Enabled 10b = Error 11b = Not available SPN 3441	Name values values values values values SPN
Lock Status Door 1 00b = Unlocked 01b = Locked 10b = Error 11b = Not available SPN 3412	Open Status Door 2 00b = Closed 01b = Open 10b = Error 11b = Not available SPN 3416	Enable Status Door 3 00b = Disabled 01b = Enabled 10b = Error 11b = Not available SPN 3420	Lock Status Door 5 00b = Unlocked 01b = Locked 10b = Error 11b = Not available SPN 3424	Open Status Door 6 00b = Closed 01b = Open 10b = Error 11b = Not available SPN 3428	Enable Status Door 7 00b = Disabled 01b = Enabled 10b = Error 11b = Not available SPN 3432	Lock Status Door 9 00b = Unlocked 01b = Locked 10b = Error 11b = Not available SPN 3436	Open Status Door 10 00b = Closed 01b = Open 10b = Error 11b = Not available SPN 3440	Name values values values values values SPN

Remark: **Lock Status:** **locked** -> doors cannot be operated by the driver or a passenger
unlocked -> door may be operated by the driver or a passenger
Enable Status: **disabled** -> door cannot be opened by a passenger
enabled -> door can be opened by a passenger
Open Status: **closed** -> door is completely closed
open -> door is not completely closed

Additional comment:
For the door configuration please contact the manufacturer of the vehicle.

1.3.5 Time / Date : TD

0x00FEE6								PGN Hex
65,254								PGN
1000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte	Data Byte	Byte No
8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1			Bit No.
Seconds 0.25 s/Bit 0 Offset SPN 959	Minutes 1 min /Bit 0 offset SPN 960	Hours 1 hr/Bit 0 offset SPN 961	Month 1 month/Bit 0 offset SPN 963	Day 0.25 day/Bit 0 offset SPN 962	Year 1 year/Bit 1985 years offset SPN 964	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Name values values values values values SPN

Description acc. SAE J 1939:

Seconds: Part of a parameter used to represent time.
Minutes: Part of a parameter used to represent time.
Hours: Part of a parameter used to represent time.
Month: Part of a parameter used to represent a calendar date.
NOTE - A value of 0 for the month is null. The value 1 identifies January; 2 identifies February; etc.
Day: Part of a parameter used to represent a calendar date.
NOTE - A value of 0 for the date is null. The values 1, 2, 3, and 4 are used to identify the first day of the month; 5, 6, 7, and 8 identify the second day of the month; etc.
Year: Part of a parameter used to represent a calendar date.
NOTE - A value of 0 for the year identifies the year 1985; a value of 1 identifies 1986; etc.

Additional comment:

Difference to SAE: broadcast instead of on request.

Time base is OEM dependant *

1.3.6 Alternator Speed : AS

0x00FED5								PGN Hex
65,237								PGN
1000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
		8 7 6 5 4 3 2 1						Bit No.
Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Alternator Status 4 00b = not charging 01b = charging 10b = error 11b = not available SPN 3356	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Name values values values values values SPN
		Alternator Status 3 00b = not charging 01b = charging 10b = error 11b = not available SPN 3355						Name values values values values values SPN
		Alternator Status 2 00b = not charging 01b = charging 10b = error 11b = not available SPN 3354						Name values values values values values SPN
		Alternator Status 1 00b = not charging 01b = charging 10b = error 11b = not available SPN 3353						Name values values values values values SPN

Description acc. SAE J 1939:

Actual alternator 1-4 status

1.3.7 Electronic Transmission Controller 2 : ETC2

0x00F005								PGN Hex
61,445								PGN
100 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 7 6 5 4 3 2 1			8 7 6 5 4 3 2 1					Bit No.
Selected Gear 1 gear value/Bit -125 offset negative gear are reverse gears 00000000b = neutral 11111011b = park SPN 524	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Current Gear 1 gear value/Bit -125 offset negative gear are reverse gears 00000000b = neutral 11111011b = park SPN 523	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Name values values values values values values values SPN

Description acc. SAE J 1939:

Selected Gear: The gear that the transmission will attempt to achieve during the current shift if a shift is in progress, or the next shift if one is pending (i.e., waiting for torque reduction to initiate the shift).

Current Gear : The gear currently engaged in the transmission or the last gear engaged while the transmission is in the process of shifting to the new or selected gear. Transitions toward a destination gear will not be indicated. Once the selected gear has been engaged then Current Gear will reflect that gear.

Additional comment:

The signal might have limited resolution for gearboxes without electronic control units *

1.3.8 Air Suspension Control 4 : ASC4

0x00FE58								PGN Hex
65,112								PGN
100 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 - 1	8 - 1	8 - 1	8 - 1	8 - 1	8 - 1	8- 1	8 - 1	Bit No.
Bellow Pressure Front Axle Left 0.1 kPa/Bit 0 offset SPN 1725	Bellow Pressure Front Axle Left 0.1 kPa/Bit 0 offset SPN 1725	Bellow Pressure Front Axle Right 0.1 kPa/Bit 0 offset SPN 1726	Bellow Pressure Front Axle Right 0.1 kPa/Bit 0 offset SPN 1726	Bellow Pressure Rear Axle Left 0.1 kPa/Bit 0 offset SPN 1727	Bellow Pressure Rear Axle Left 0.1 kPa/Bit 0 offset SPN 1727	Bellow Pressure Rear Axle Right 0.1 kPa/Bit 0 offset SPN 1728	Bellow Pressure Rear Axle Right 0.1 kPa/Bit 0 offset SPN 1728	Name Name values values values values SPN

Description acc. SAE J 1939:

Used for bellow pressure information

Additional comment:

The configuration is manufacturer dependant.*

The mentioned resolution is not related to the accuracy of the signal.

1.3.9 Vehicle Electrical Power #4 : VEP4

0x00FCB7								PGN Hex
64,695								PGN
10000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 - 1	8 - 1	8 - 1	8 - 1	8 - 1	8 - 1	8- 1	8 - 1	Bit No.
Hybrid Battery Pack Remaining Charge 0.0025 %/bit 0 to 160.6375 % SPN 5464	Hybrid Battery Pack Remaining Charge 0.0025 %/bit 0 to 160.6375 % SPN 5464	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Name Name values values values values SPN

Description acc. SAE J 1939:

Indicates the hybrid battery pack remaining charge.
0% means no charge remaining,
100% means full charge remaining.

Additional comment:

The mentioned resolution is not related to the accuracy of the signal.

This SPN is used as well for electric vehicles not only for hybrid.

1.3.10 Vehicle Dynamic Stability Control 2 : VDC2

0x00F009								PGN Hex
61,449								PGN
100 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 - 1	8 - 1	8 - 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	Bit No.
Steering Wheel Angle 1/1024 rad per bit, -31.374 rad offset SPN 1807	Steering Wheel Angle 1/1024 rad per bit, -31.374 rad offset SPN 1807	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Lateral Acceleration 1/2048 m/s² per bit -15.687 m/s² offset SPN 1809	Lateral Acceleration 1/2048 m/s² per bit -15.687 m/s² offset SPN 1809	Longitudinal Acceleration 0.1 m/s² per bit -12.5 m/s² offset SPN 1810	Name Name values values values values SPN

Description acc. SAE J 1939:

Steering wheel angle: The main operator`s steering wheel angle (on the steering column, not the actual wheel angle).
The vehicle being steered to the left (counterclockwise) results in a positive steering wheel angle.
This is the yaw angle of the steering wheel with the z-axis along the centerline of the steering column.
This parameter is defined according to a Z-Up axis system and the sign of the value is in accordance to the right-hand rule, as specified in SAE J670.
As specified in SAE J670, a Z-Up Axis System has positive X directed forward, positive Y to the left, and positive Z directed up
Lateral Acceleration: Indicates a lateral acceleration of the vehicle (the component of vehicle acceleration vector along the Y-axis). A positive lateral acceleration signal results when the vehicle is accelerated to the left. This parameter is defined according to a Z-Up axis system and the sign of the value is in accordance to the right-hand rule, as specified in SAE J670. As specified in SAE J670, a Z-Up Axis System has positive X directed forward, positive Y to the left, and positive Z directed up. See SPN 5347 for an alternate range and resolution.
Longitudinal Acceleration: Indicates the longitudinal acceleration of the vehicle. A positive longitudinal acceleration signal results when the vehicle speed increases, regardless of driving the vehicle forward or backward. This parameter is applicable to vehicle speed measurement systems, such as passive wheel speed sensors, that are not capable of determining the direction of the vehicle motion. See SPN 5348 for an alternate longitudinal acceleration.

Additional Comment:

The mentioned resolution is not related to the accuracy of the signal.

The repetition rate can be higher than 100ms and is OEM dependant.*

1.3.11 Engine Temperature 1: ET1

0x00FEEE								PGN Hex
65,262								PGN
1000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
8 7 6 5 4 3 2 1		8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1					Bit No
Engine coolant temperature 1 °C / Bit gain - 40 °C offset SPN 110	Not used for (Bus) FMS-Standard	Engine Oil Temperature 1 0.03125 °C / Bit - 273 °C offset SPN 175	Engine Oil Temperature 1 0.03125 °C / Bit - 273 °C offset SPN 175	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Name values values values SPN

Description acc. SAE J 1939:

Engine Coolant Temperature: Temperature of liquid found in engine cooling system.
Engine Oil Temperature: Temperature of the engine lubricant.

Additional comment:
The mentioned resolution is not related to the accuracy of the signal.

1.3.12 Ambient Conditions: AMB

0x00FEF5								PGN Hex
65,269								PGN
1000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1				Bit No
Not used for (Bus) FMS-Standard	Cab interior temperature 0.03125 °C / Bit gain - 273 °C offset SPN 170	Cab interior temperature 0.03125 °C / Bit gain - 273 °C offset SPN 170	Ambient Air Temperature 0.03125 °C / Bit gain - 273 °C offset SPN 171	Ambient Air Temperature 0.03125 °C / Bit gain - 273 °C offset SPN 171	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Not used for (Bus) FMS-Standard	Name Name values values values values SPN

Description acc. SAE J 1939:

Ambient Air Temperature: Temperature of air surrounding vehicle.
Cab interior temperature: Temperature of air inside the part of the vehicle that encloses the driver and vehicle operating controls.

Additional comment:
It is not the temperature in the passenger compartment.

The mentioned resolution is not related to the accuracy of the signal.

1.3.13 High Voltage Energy Storage System Thermal Management System Status 1 : HVESSTS1

0x00F109								PGN Hex
61,705								PGN
100 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
						8 7 6 5 4 3 2 1		Bit No.
Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	HVESS Thermal Management System Mode 0000b = Off 0001b = Circulating 0010b = Cooling 0011b = Heating 0100b = Drying 0101b = Reserved 0110b = Reserved 0111b = Reserved 1000b = Reserved 1001b = Reserved 1010b = Reserved 1011b = Reserved 1100b = Reserved 1101b = Reserved 1110b = Faulted 1111b = Not available SPN 8235	Not used in Bus FMS-Standard	Name Name Name values values values values values values values values values values values values values values values values SPN

Description acc. SAE J 1939:

HVESS Thermal Management System Mode: Indicates the operating mode of the Thermal Management System. The heat transfer medium may be (e.g. liquid or air). The cooling system may be fans, liquid coolant, or a combination of mediums. The system may include an air and/or liquid conditioning system with active Cooling, Heating, and/or Drying capabilities.

Additional Comment:

The mentioned resolution is not related to the accuracy of the signal.

1.3.14 High Voltage Energy Storage System History : HVESSHIST

0x00FC5E								PGN Hex
64,606								PGN
10000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
			8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1		Bit No.
Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	HVESS Total Energy Throughput 100 Watt/bit 0 Watt offset SPN 8211	HVESS Total Energy Throughput 100 Watt/bit 0 Watt offset SPN 8211	HVESS Total Energy Throughput 100 Watt/bit 0 Watt offset SPN 8211	HVESS Total Energy Throughput 100 Watt/bit 0 Watt offset SPN 8211	Not used in Bus FMS-Standard	Name Name values values values SPN

Description acc. SAE J 1939:

HVESS Total Energy Throughput: The total energy throughput which has flowed through High Voltage Energy Storage System (or Pack 1 in systems with only one pack). This parameter is the sum of the absolute values of the Energy into and out of this HVES instance.

This parameter provides HVES system level information. If there are multiple packs in the system, then both individual pack messages (e.g., HVESP1, HVESP2 etc.) and system level messages (HVESS) may be required.

Additional Comment:

This PGN is not sent as a BAM.
The mentioned resolution is not related to the accuracy of the signal.

1.3.15 Hybrid System Status 1 : HSS1

0x00FCC2								PGN Hex
64,706								PGN
Every 1 s and on change but no faster than 100 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
	8 7 6 5 4 3 2 1					8 7 6 5 4 3 2 1		Bit No.
Not used in Bus FMS-Standard	Propulsion System Active 00b = Propulsion System is not ready to move vehicle if commanded by operator 01b = Propulsion System is ready to move, or is moving the vehicle as commanded by operator 10b = SAE Reserved 11b = Not available SPN 7315	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Name Name values values values values values values values values values values values values SPN

Description acc. SAE J 1939:

Propulsion System Active: Indicates when the hybrid system or electric vehicle system is ready to provide or is actively providing propulsion as defined in ARB regulation 13CCR1971.1, which includes the amendments up to the Board hearing date of August 2012. This parameter is called out by the 1971.1 regulation in several areas including, but not limited to: the ignition cycle counter [1971.1 paragraphs (d)(5.5.2)(B) and (d)(5.5.2)(C)], permanent fault code erasure [(d)(2.3.1)(C)(ii)b.3.v], and denominator incrementing [(d)(4.3.2)(J)(i)] requirements. For example, "for hybrid vehicles, 'ignition cycle' means a driving cycle that begins when the propulsion system active definition is met for at least two seconds plus or minus one second, and ends when the propulsion system active definition is no longer met."

NOTE: The parameter name contains the word "active" so that it matches the naming in the ARB regulation. However, this parameter not only indicates that the hybrid system or electric vehicle system is actively aiding or retarding vehicle motion (i.e., generating any positive or negative torque on the driveline); it also shall be asserted when the hybrid system or powertrain system is ready to participate in such activities if vehicle conditions or operator input warrant. Therefore, this parameter is asserted when the vehicle is ready to provide propulsion (e.g., vehicle is ready to be driven, ready to be shifted from "park" to "drive") and doesn't change state again until the system is neither ready to provide, nor actively providing, propulsion.

This parameter is required for hybrid vehicles because there is a difference between a conventional vehicle, where powertrain readiness is equivalent to "Engine Running", and a hybrid vehicle, where the engine may or may not be running and yet the vehicle could move. However, if the engine is running (engine speed is nonzero), this parameter shall be set to 1. If the engine speed is zero, then this parameter shall be set based on the hybrid system operating state. For heavy duty vehicles, the hybrid system needs to inform the OBD related components of its propulsion active or readiness state to allow the components to perform OBD related functions.

Propulsion System Active cannot be used as an indication of propulsion availability as it is required to indicate "active" whenever the engine in a hybrid system is running. This is regardless of the hybrid system's ability or powertrain system's ability to provide propulsion power. See SPN 8267 for an indication of hybrid system propulsion availability.

This message may be used by the Hybrid System Controller to communicate to other OBD related components (e.g engine) the status of the overall hybrid system. It should not be sent by individual hybrid system components (e.g., batteries, inverters, etc.)

This message may also be used on electric vehicles to communicate the status of the powertrain system on the electric vehicle.

Indicates when the hybrid system or electric vehicle system is ready to provide or is actively providing propulsion.

Additional Comment:

The mentioned resolution is not related to the accuracy of the signal

1.3.16 Engine Fluid Level/Pressure 1 : EFL/P1

0x00FEEF								PGN Hex
65,263								PGN
500 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
			8 7 6 5 4 3 2 1					Bit No.
Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Engine Oil Pressure 1 4 kPa/bit 0 offset SPN 100	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Name values values SPN

Description acc. SAE J 1939:

Engine Oil Pressure 1: Gage pressure of oil in engine lubrication system as provided by oil pump.

Additional Comment:

The mentioned resolution is not related to the accuracy of the signal.

1.3.17 Transmission Fluids 1 : TRF1

0x00FEF8								PGN Hex
65,272								PGN
1000 ms								Rep. Rate
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4	Data Byte 5	Data Byte 6	Data Byte 7	Data Byte 8	Byte No
				8 7 6 5 4 3 2 1	8 7 6 5 4 3 2 1			Bit No.
Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Transmission Oil Temperature 1 0.03125 °C/bit -273 °C offset SPN 177	Transmission Oil Temperature 1 0.03125 °C/bit -273 °C offset SPN 177	Not used in Bus FMS-Standard	Not used in Bus FMS-Standard	Name Name values values SPN

Description acc. SAE J 1939:

Transmission Oil Temperature 1: First instance of transmission lubricant temperature.

Additional Comment:

The mentioned resolution is not related to the accuracy of the signal.

2 Examples

2.1 Broadcast Announce Message (BAM) for Vehicle ID longer than 8 Byte

Transport Protocol – Connection Management (TP.CM)

0x00ECFF								PGN Hex
60,671								PGN
Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte No
								Bit No
Control byte should be filled with (20_{16})	Total message size, number of bytes	Total message size, number of bytes	Total number of packets	Reserved should be filled with FF_{16}	Parameter Group Number of packeted message	Parameter Group Number of packeted message	Parameter Group Number of packeted message	Name values values values SPN

Transport Protocol – Data Transfer (TP.DT)

0x00EBFF								PGN Hex
60,415								PGN
Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte No
								Bit No
Sequence Number	Packetized Data	Packetized Data	Packetized Data	Packetized Data	Packetized Data	Packetized Data	Packetized Data	Name values values values SPN

In the situation shown in Figure 1, a node indicates to the network that it is about to transfer a multipacket message utilizing the service of the transport protocol.
In this example, the PGN $00FEEC_{16}$ (Vehicle Identification) is being broadcasted to the network.
The length of the Vehicle ID in this example is 17.
The unused bytes in the last TP.DT are filled with FF_{16}.
The originating node first transmits a TP.CM Broadcast Announce Message (BAM) followed by the data packets.
No acknowledgment is performed by any of the responders.

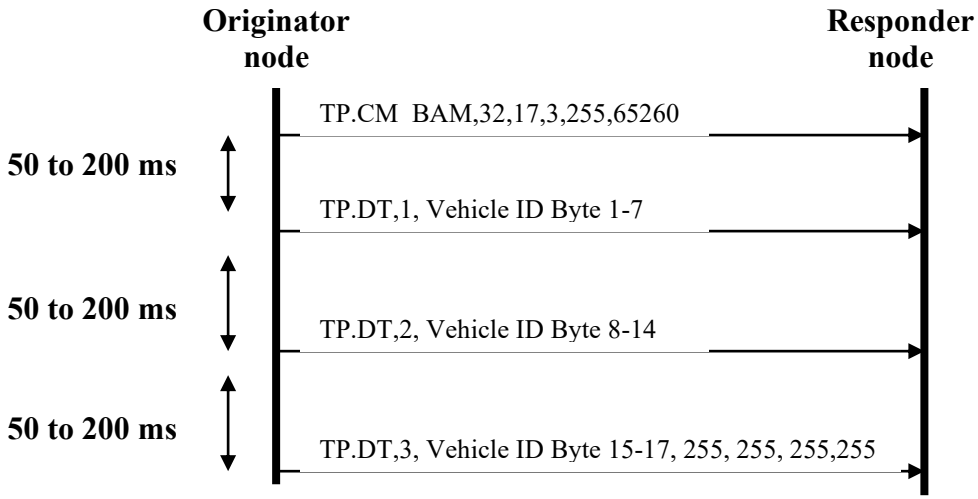

Figure 1

Time (ms)	ID	DLC	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
0	PR EC FF SA $_{16}$	8	20 $_{16}$	11 $_{16}$	00 $_{16}$	03 $_{16}$	FF $_{16}$	EC $_{16}$	FE $_{16}$	00 $_{16}$
50	PR EB FF SA $_{16}$	8	01 $_{16}$	Vehicle ID byte 1 – 7						
100	PR EB FF SA $_{16}$	8	02 $_{16}$	Vehicle ID byte 8 – 14						
150	PR EB FF SA $_{16}$	8	03 $_{16}$	Vehicle ID byte 15	Vehicle ID byte 16	Vehicle ID byte 17	FF $_{16}$	FF $_{16}$	FF $_{16}$	FF $_{16}$

PR is Priority (to be masked)
SA is Source Address (to be masked)

2.2 Example SW Identification for buses and/or trucks

	ID	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
Supporting Bus-FMS-Standard Version 01	0x00FD D1 $_{16}$	X0 $_{16}$	30 $_{16}$	30 $_{16}$	30 $_{16}$	31 $_{16}$	FF $_{16}$	FF $_{16}$	FF $_{16}$
Supporting Truck-FMS-Standard Version 02	0x00FD D1 $_{16}$	X0 $_{16}$	30 $_{16}$	32 $_{16}$	30 $_{16}$	30 $_{16}$	FF $_{16}$	FF $_{16}$	FF $_{16}$
Supporting Bus FMS-Standard Version 03 and Truck FMS-Standard Version 03	0x00FD D1 $_{16}$	X0 $_{16}$	30 $_{16}$	33 $_{16}$	30 $_{16}$	33 $_{16}$	FF $_{16}$	FF $_{16}$	FF $_{16}$

Remark: **Byte 2 – Byte 5 are ASCII** **X=reserved and set to F_{16}**

30_{16} = “0” ASCII
31_{16} = “1” ASCII
32_{16} = “2” ASCII
33_{16} = “3” ASCII

2.3 Example FMS Tell Tale Status

ID	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
0x00FD7D $_{16}$	**B0 $_{16}$**	**FF $_{16}$**	**FF $_{16}$**	**9F $_{16}$**	**FA $_{16}$**	**A8 $_{16}$**	**9B $_{16}$**	**FF $_{16}$**
TTS	Block ID = 0 Status 1 = Info	Status 2 = not av. Status 3 = not av.	Status 4= not av. Status 5= not av.	Status 6= not av. Status 7= Red	Status 8= Yellow Status 9= not av.	Status 10= off Status 11= Yellow	Status 12= Info Status 13= Red	Status 14= not av. Status 15= not av.
0x00FD7D $_{16}$	**F1 $_{16}$**	**8F $_{16}$**	**AF $_{16}$**	**B8 $_{16}$**	**FB $_{16}$**	**88 $_{16}$**	**AA $_{16}$**	**BB $_{16}$**
TTS	Block ID = 1 Status 1 = not av.	Status 2 = not av. Status 3 = off	Status 4= not av. Status 5= Yellow	Status 6= off Status 7= Info	Status 8= Info Status 9= not av.	Status 10= off Status 11= off	Status 12= Yellow Status 13= Yellow	Status 14= Info Status 15= Info
0x00FD7D $_{16}$	**A2 $_{16}$**	**F8 $_{16}$**	**88 $_{16}$**	**FF $_{16}$**	**FF $_{16}$**	**AF $_{16}$**	**FF $_{16}$**	**FF $_{16}$**
TTS	Block ID = 2 Status 1 = Yellow	Status 2 = off Status 3 = not av.	Status 4= off Status 5= off	Status 6= not av. Status 7= not av.	Status 8= not av. Status 9= not av.	Status 10= not av. Status 11= Yellow	Status 12= not av. Status 13= not av.	Status 14= not av. Status 15= not av.
0x00FD7D $_{16}$	**F3 $_{16}$**	**BB $_{16}$**	**88 $_{16}$**	**88 $_{16}$**	**8B $_{16}$**	**88 $_{16}$**	**88 $_{16}$**	**88 $_{16}$**
TTS	Block ID = 3 Status 1 = not av.	Status 2 = Info Status 3 = Info	Status 4= off Status 5= off	Status 6= off Status 7= off	Status 8= Info Status 9= off	Status 10= off Status 11= off	Status 12= off Status 13= off	Status 14= off Status 15= off

Remark: **The repetition rate of the PGN is 1000ms which means that the complete "Tell Tale" Msg. (five Blocks) is sent every 5000 ms**

Due to the repetition rate of the message, it is not guaranteed to include all intermittent signals.*

not av. = not available

Block ID	Status No	Status	Name		Block ID	Status No	Status	Name
0	1	Info	Cooling air conditioning		2	1	Yellow	Steering failure
0	2	Not av.	High beam, main beam		2	2	Off	Height Control (Levelling)
0	3	Not av.	Low beam, dipped beam		2	3	Not av.	Retarder
0	4	Not av.	Turn signals		2	4	Off	Engine Emission system failure (Mil indicator)
0	5	Not av.	Hazard warning		2	5	Off	ESC indication
0	6	Not av.	Provision for the disabled or handicapped persons		2	6	Not av.	Brake lights
0	7	Red	Parking Brake		2	7	Not av.	Articulation
0	8	Yellow	Brake failure/brake system malfunction		2	8	Not av.	Stop Request
0	9	Not av.	Hatch open		2	9	Not av.	Pram request
0	10	Off	Fuel level		2	10	Not av.	Bus stop brake
0	11	Yellow	Engine coolant temperature		2	11	Yellow	AdBlue level
0	12	Info	Battery charging condition		2	12	Not av.	Raising
0	13	Red	Engine oil		2	13	Not av.	Lowering
0	14	Not av.	Position lights,side lights		2	14	Not av.	Kneeling
0	15	Not av.	Front fog light		2	15	Not av.	Engine compartment temperature
1	1	Not av.	Rear fog light		3	1	Not av.	Auxillary air pressure
1	2	Not av.	Park Heating		3	2	Info	Air filter clogged
1	3	Off	Engine		3	3	Info	Fuel filter differential pressure
1	4	Not av.	Service, call for maintenance		3	4	Off	Seat belt
1	5	Yellow	Transmission fluid temperature		3	5	Off	EBS
1	6	Off	Transmission failure/malfunction		3	6	Off	Lane departure indication
1	7	Info	Anti-lock brake system failure		3	7	Off	Advanced emergency braking system
1	8	Info	Worn brake linings		3	8	Info	ACC
1	9	Not av.	Windscreen washer fluid/windshield washer fluid		3	9	Off	Trailer connected
1	10	Off	Tire failure/malfunction		3	10	Off	ABS Trailer 1,2
1	11	Off	Malfunction/general failure		3	11	Off	Airbag
1	12	Yellow	Engine oil temperature		3	12	Off	EBS Trailer 1,2
1	13	Yellow	Engine oil level		3	13	Off	Tachograph indication
1	14	Info	Engine coolant level		3	14	Off	ESC switched off
1	15	Info	Steering fluid level		3	15	Off	Lane departure warning switched off

3 Overview Messages

			(signal) name	Mandatory	rep. rate	remarks / comments	
page	PGN	SPN	e.g. milage, fuel consumption	Truck only	in ms		
Truck AND Bus Section			Truck AND Bus Section	Truck AND Bus Section		Truck AND Bus Section Truck AND Bus Section	Truck AND Bus Section Truck AND Bus Section
11	65257	250	Engine total fuel used		1000	4 bytes, 0 to +2 105 540 607,5 L	Might be set to “not available” if SPN 5054 is available
12	65276	96	Fuel level 1	X (worldwide)	1000	1 Byte	-
12	65276	38	Fuel level 2		1000	1 Byte	Only used if different fuel types in different storage container
13	61444	513	Actual Engine – Percent Torque	X (worldwide)	20	1 % / Bit, -125 % offset	-
13	61444	190	Engine speed	X (worldwide)	20	2 Byte, 0-8031,875 rpm	-
14	65253	247	Engine total hours of Operation	X (worldwide)	1000	4 bytes, 0 to 210 554 060,75 h	Counter is Engine running dependant
15	65260	237	vehicle identification number	X (worldwide)	10000	variable, max 200 char.	Will be sent every 10 sec
16	64977	2806	SW-version supported	X (worldwide)	10000	Indicator for SW version supported	-
16	64977	2804	Diagnostics supported	X (worldwide)	10000	indicator for diagnosis session support	-
16	64977	2805	Requests supported	X (worldwide)	10000	indicator for request supported	-
18	65217	917	High resolution total vehicle distance	X (worldwide)	1000	4 bytes, 0 - 21 055 406 km; without TCO	Resolution may be not within the SAE values
19	65132	1611	Vehicle motion	X (EU)	20/50	With digital tachograph	rep. rate tacho dependant
19	65132	1613	Driver 2 working state	X (EU)	20/50	With digital tachograph	rep. rate tacho dependant
19	65132	1612	Driver 1 working state	X (EU)	20/50	With digital tachograph	rep. rate tacho dependant
19	65132	1614	Vehicle overspeed		20/50	With digital tachograph	rep. rate tacho dependant
19	65132	1617	Driver 1 time rel. states		20/50	With digital tachograph	rep. rate tacho dependant
19	65132	1618	Driver 2 time rel. states		20/50	With digital tachograph	rep. rate tacho dependant
19	65132	1615	Driver 1 card	X (EU)	20/50	With digital tachograph	rep. rate tacho dependant
19	65132	1616	Driver 2 card	X (EU)	20/50	With digital tachograph	rep. rate tacho dependant
19	65132	1619	Direction indicator		20/50	With digital tachograph	rep. rate / availability is tachograph dependant.
19	65132	1620	Tachograph performance	X (EU)	20/50	With digital tachograph	rep. rate tacho dependant
19	65132	1621	Handling information	X (EU)	20/50	With digital tachograph	rep. rate tacho dependant
19	65132	1622	System event	X (EU)	20/50	With digital tachograph	rep. rate tacho dependant
19	65132	1624	Tachograph vehicle speed	X (EU)	20/50	With digital tachograph - 2 bytes	rep. rate tacho dependant/might differ from the wheel based speed
21	65262	110	Engine coolant temperature	X (worldwide)	1000	-40° to 210°	-
22	65269	171	Ambient Air Temperature	X (worldwide)	1000	0.03125 °C / Bit gain	- 273 °C offset
23	65131	1625/1626	Driver 1 / Driver 2 Identification	X (EU)	10000	If a driver ID is available the message is sent with a Broadcast Announce Message (BAM)	Diff. to SAE: broadcast instead of on request
24	65266	183	Fuel rate	X (worldwide)	100	0.05 L/h per bit, 0 to 3,212.75 L/h	Calculated values given as indications, not as contractual
24	65266	184	Instantaneous Fuel Economy	X (worldwide)	100	1/512 km/L per bit, 0 to 125,5 km/L	Calculated values given as indications, not as contractual
25	65198	1087	Service Brake Air Pressure Circuit #1	X (worldwide)	1000	8 kPa/Bit, 0 offset	-
25	65198	1088	Service Brake Air Pressure Circuit #2	X (worldwide)	1000	8 kPa/Bit, 0 offset	-
26	64777	5054	High resolution engine total fuel used		1000	0.001 L/bit, 0 to 4,211,081.215 L	Is implemented if technical possible
27	65110	1761	Aftertreatment 1 Diesel Exhaust Fluid Tank 1 Level		1000	0.4 %/bit, 0 % offset	
28	64893		FMS Tell Tale Status	X (worldwide ,EU) Not all tell tales	1000	-	5 blocks => Rep.rate for each tell tale status might be 5 sec if all blocks are sent
32	61441	521	Brake Pedal Position		100	-	-
33	64962	5837	Fuel Type		10000	-	-
36	65199	1040	Total fuel used (Gaseous)		1000	-	-
37	61440	900	Retarder Torque Mode		100	16 states/4 bit, 0 offset	-
37	61440	520	Actual Retarder - Percent Torque		100	1 %/bit, -125 % offset	-
37	61440	1716	Retarder Selection, non-engine		100	0.4 %/bit, 0 % offset	The value is related to the drive line retarder
38	65258	928	Axle location		1000	-	If info of more axles available it will be updated with each repetition
38	65258	928	Tire location		1000	-	-
38	65258	582	Axle weight		1000	-	-
39	64184	13171	DC Charging State		1000	-	-
40	64617	15268	HVESS Estimated Remaining Distance		10000	-	-
41	65268	929	Tire Location		10000	-	-
41	65268	241	Tire Pressure		10000	-	-
42	65279	97	Water In Fuel Indicator 1		10000	-	-
42	65279	8428	Fuel Supply Estimated Remaining Distance		10000	-	SPN 15268 is used for BEV instead of SPN 8428

Truck only Section			Truck only Section	Truck only Section		Truck only Section	Truck only Section	Truck only Section	Truck only Section
43	65265	84	Wheel based speed		100	may differ from TCO1		-	
43	65265	598	Clutch switch		100	two bit status		in trucks with automatic gear => send as not available	
43	65265	597	Brake switch		100	two bit status		-	
43	65265	595	Cruise control active		100	two bit status		in trucks with no cruise control => send as not available	
43	65265	976	PTO state		100	Either SPN 3948 (PTODE) or SPN 976 is sent		SPN 3948 (PTO DE) message is preferred	
43	65265	70	Parking brake switch		100	-		-	
43	65265	527	Cruise Control States		100	-		-	
45	61443	91	Accelerator pedal position 1	X (worldwide)	50	1 Byte		-	
45	61443	92	Engine Percent Load At Current Speed	X (worldwide)	50	1 % / bit, 0 to 125 % operational range		-	
46	65216	914	Service distance		1000	-		-	
47	64932	3948	At least one PTO engaged		100	Either SPN 3948 or SPN 976 (CCVS) is sent		SPN 3948 (PTO DE) message is preferred	
48	65136	1760	Gross Combination Vehicle Weight		10000	0 to 642,550 kg		Diff. to SAE: broadcast instead of on request	
Bus only Section			**Bus only Section**	**Bus only Section**		**Bus only Section**	**Bus only Section**	**Bus only Section**	**Bus only Section**
49	65265	84	Wheel based speed		100	may differ from TCO1		-	
49	65265	598	Clutch switch		100	two bit status		in trucks with automatic gear => send as not available	
49	65265	597	Brake switch		100	two bit status		-	
49	65265	595	Cruise control active		100	two bit status		in trucks with no cruise control => send as not available	
49	65265	70	Parking Brake Switch		100	-		-	
49	65265	527	Cruise Control States		100	-		-	
51	61443	91	Accelerator pedal position 1		50	1 Byte		-	
52	65102	3411	Status 2 of doors		100	-		-	
52	65102	1820	Ramp/wheel chairlift status		100	-		-	
52	65102	1821	Position of doors		100	-		For door configuration please contact the vehicle manufacturer	
53	64933	3412-3441	Status of doors 1 - 10		100	-		For door configuration please contact the vehicle manufacturer	
54	65254	959	Seconds		1000	-		Difference to SAE: broadcast instead of on request	
54	65254	960	Minutes		1000	-		Difference to SAE: broadcast instead of on request	
54	65254	961	Hours		1000	-		Difference to SAE: broadcast instead of on request	
54	65254	963	Month		1000	-		Difference to SAE: broadcast instead of on request	
54	65254	962	Day		1000	-		Difference to SAE: broadcast instead of on request	
54	65254	964	Year		1000	-		Difference to SAE: broadcast instead of on request	
55	65237	3356	Alternator Status 4		1000	-		-	
55	65237	3355	Alternator Status 3		1000	-		-	
55	65237	3354	Alternator Status 2		1000	-		-	
55	65237	3353	Alternator Status 1		1000	-		-	
56	61445	524	Selected Gear		100	-		-	
56	61445	523	Current Gear		100	-		-	
57	65112	1725	Bellow Pressure Front Axle Left		100	-		-	
57	65112	1726	Bellow Pressure Front Axle Right		100	-		-	
57	65112	1727	Bellow Pressure Rear Axle Left		100	-		-	
57	65112	1728	Bellow Pressure Rear Axle Right		100	-		-	
58	64695	5464	Hybrid Battery Pack Remaining Charge		10000	-		This SPN is used as well for electric vehicles not only for hybrid.	
59	61449	1807	Steering wheel angle		100	-		-	
59	61449	1809	Lateral Acceleration		100	-		-	
59	61449	1810	Longitudinal Acceleration		100	-		-	
60	65262	110	Engine coolant temperature		1000	-		-	
60	65262	175	Engine Oil Temperature 1		1000	-		-	
61	65269	170	Cab interior temperature		1000	-		It is not the temperature in the passenger compartment.	
61	65269	171	Ambient Air Temperature		1000	-		-	
62	61705	8235	HVESS Thermal Management System Mode		100	-		-	
63	64606	8211	HVESS Total Energy Throughput		10000	-		-	
64	64706	7315	Propulsion System Active		100	-		-	
66	65263	100	Engine Oil Pressure 1		500	-		-	
67	65272	177	Transmission Oil Temperature 1		1000	-		-	